King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 102 Final Exam Term 243

Original exam date: 5 August 2025. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. The coefficient of x^9 in the Maclaurin series for the function $f(x) = 2\sin x^3$ is

(a) $\frac{1}{2}$

(b) $-\frac{1}{2}$

(c) $\frac{1}{3}$

(d) $\frac{1}{6}$

(e) $-\frac{1}{3}$

2. Which one of the following series is/are convergent?

$$\text{I)} \sum_{n=1}^{\infty} \frac{(n-1)!}{4^n} \qquad \text{II)} \sum_{n=1}^{\infty} n^2 \left(\frac{5}{6}\right)^n \qquad \text{III)} \sum_{n=1}^{\infty} \left(\frac{n-2}{5n+1}\right)^n$$

(a) I and II only

(b) II and III only

(c) I and III only

(d) III only

(e) II only

3. The volume of the solid generated by revolving the region bounded by the graphs of $y = 2x^2$, $y = 0$, and $x = 2$ about the x-axis is

(a) $\frac{16\pi}{5}$

(b) $\frac{128\pi}{5}$

(c) $\frac{8\pi}{5}$

(d) $\frac{64\pi}{5}$

(e) $\frac{32\pi}{5}$

4.

$$\int_0^2 x^2 e^{-2x}\,dx =$$

(a) $\dfrac{13+e^4}{4e^4}$

(b) $\dfrac{-13-e^4}{4e^4}$

(c) $\dfrac{-13+e^4}{e^4}$

(d) $\dfrac{13-e^4}{4e^4}$

(e) $\dfrac{-13+e^4}{4e^4}$

5.

$$\sum_{n=0}^{\infty}[(0.3)^n+(0.8)^n] =$$

(a) $\dfrac{45}{7}$

(b) $\dfrac{43}{7}$

(c) $\dfrac{44}{7}$

(d) $\dfrac{47}{7}$

(e) $\dfrac{46}{7}$

6. The sequence whose nth term is $a_n = \dfrac{\sin n}{n}$ converges to

(a) π

(b) 0

(c) 2

(d) $\dfrac{\pi}{2}$

(e) 2π

7.

$$\int_0^2 \frac{dx}{\sqrt[3]{x-1}}$$

(a) converges to 3

(b) diverges

(c) converges to 2

(d) converges to 1

(e) converges to 0

8. Which one of the following series is/are convergent?

$$\text{I)} \sum_{n=1}^{\infty} \frac{\sin^2 n}{n^3} \qquad \text{II)} \sum_{n=1}^{\infty} \frac{6^n + n}{5^n - 1} \qquad \text{III)} \sum_{n=1}^{\infty} \frac{n}{(n+1)2^{n-1}}$$

(a) I only

(b) II only

(c) I and II only

(d) II and III only

(e) I and III only

9. A power series representation for the function $f(x) = \arctan(2x)$ centered at $c = 0$ is

(a) $2\sum_{n=0}^{\infty}(-1)^n 4^n x^{2n+1}$

(b) $2\sum_{n=0}^{\infty}\frac{(-1)^n 2^n x^{2n+1}}{2n+1}$

(c) $\sum_{n=0}^{\infty}\frac{(-1)^n 4^n x^{2n+1}}{2n+1}$

(d) $\sum_{n=0}^{\infty}\frac{(-1)^n 2^n x^{2n+1}}{2n+1}$

(e) $2\sum_{n=0}^{\infty}\frac{(-1)^n 4^n x^{2n+1}}{2n+1}$

10. The area of the region bounded by the graphs of the equations

$$y = \frac{-4}{x^3},\ y = 0,\ x = -3 \text{ and } x = -1$$

is

(a) $\frac{13}{9}$

(b) $\frac{16}{9}$

(c) $\frac{14}{9}$

(d) $\frac{17}{9}$

(e) $\frac{19}{9}$

11.

$$\int_0^2 \frac{3}{4x^2 + 5x + 1}\, dx =$$

(a) $\ln 3$

(b) $\ln 5$

(c) $\ln 2$

(d) $\ln 6$

(e) $\ln 4$

12.

$$\int_1^4 \frac{2x + 1}{2x}\, dx =$$

(a) $2 + \ln 3$

(b) $2 - \ln 3$

(c) $3 - \ln 2$

(d) $3 + \ln 3$

(e) $3 + \ln 2$

13. Which one of the following series is/are divergent?

$$\text{I)} \sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}} \qquad \text{II)} \sum_{n=1}^{\infty} e^{-n} \qquad \text{III)} \sum_{n=1}^{\infty} \frac{1}{n^{\pi}}$$

(a) I and III only

(b) II and III only

(c) I only

(d) I, II and III

(e) I and II only

14. The series $\sum_{n=0}^{\infty} \frac{\cos n\pi}{n+1}$

(a) diverges by the Alternating Series Test

(b) converges conditionally

(c) converges absolutely

(d) diverges

(e) converges absolutely by the Integral Test

15.

$$\int \frac{dx}{e^{2x} + e^{-2x}} =$$

(a) $\arctan(e^{-2x}) + C$

(b) $\arctan(e^{2x}) + C$

(c) $\frac{1}{2}\arctan(e^{2x}) + C$

(d) $\frac{1}{2}\arctan(e^{-2x}) + C$

(e) $\ln(e^{2x} + e^{-2x}) + C$

16.

$$\int e^x \sqrt{1 - e^{2x}}\, dx =$$

(a) $\frac{1}{2}\left(\arccos(e^x) + e^x\sqrt{1 - e^{2x}}\right) + C$

(b) $\arcsin(e^x) + e^x\sqrt{1 - e^{2x}} + C$

(c) $\frac{1}{2}\left(\arcsin(e^x) + e^x\sqrt{1 - e^{2x}}\right) + C$

(d) $\frac{1}{2}\left(\arcsin(e^x) - e^x\sqrt{1 - e^{2x}}\right) + C$

(e) $\arccos(e^x) + e^x\sqrt{1 - e^{2x}} + C$

17. The interval of convergence for the series $\sum_{n=1}^{\infty} \frac{(x-3)^{n+1}}{(n+1)4^{n+1}}$ is

(a) $[-1, 7]$

(b) $(-\infty, \infty)$

(c) $[-1, 7)$

(d) $(-1, 7]$

(e) $(-1, 7)$

18.

$$\int_0^{\infty} \frac{e^x}{1+e^x}\, dx$$

(a) diverges

(b) converges to e

(c) converges to $\dfrac{1}{e}$

(d) converges to 1

(e) converges to 0

19. Using a third degree Maclaurin polynomial of the function $f(x) = \tan x$, the approximation of the value $\tan 1$ is equal to

(a) $\dfrac{7}{3}$

(b) $\dfrac{5}{3}$

(c) $\dfrac{2}{3}$

(d) $\dfrac{8}{3}$

(e) $\dfrac{4}{3}$

20. The least number of terms required to approximate the sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$ with an error of less than 0.001 is (Hint: Use the Remainder Theorem)

(a) 12

(b) 10

(c) 18

(d) 14

(e) 16

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(e)**	$2\sin x^3 = 2x^3 - \frac{2x^9}{6} + \cdots$, coefficient $-\frac{1}{3}$.
2	**(b)**	I: ratio $n/4 \to \infty$; II: ratio $\to 5/6$; III: root $\to 1/5$. II and III converge.
3	**(b)**	Disks: $V = \pi \int_0^2 4x^4 dx = \frac{128\pi}{5}$.
4	**(e)**	Parts twice: $\left[-\frac{e^{-2x}}{4}(2x^2+2x+1)\right]_0^2 = \frac{-13+e^4}{4e^4}$.
5	**(a)**	$\frac{1}{0.7} + \frac{1}{0.2} = \frac{10}{7} + 5 = \frac{45}{7}$.
6	**(b)**	$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$, squeeze: limit 0.
7	**(e)**	Split at $x = 1$: $-\frac{3}{2} + \frac{3}{2} = 0$, converges.
8	**(e)**	I: $\le 1/n^3$; II: terms $\to \infty$; III: $\le (1/2)^{n-1}$. I and III converge.
9	**(e)**	Substitute $u = 2x$ in $\arctan u$: $2\sum \frac{(-1)^n 4^n x^{2n+1}}{2n+1}$.
10	**(b)**	$y > 0$ on $[-3,-1]$: $\int \frac{-4}{x^3} dx = [2/x^2]_{-3}^{-1} = \frac{16}{9}$.
11	**(a)**	$\frac{3}{(4x+1)(x+1)} = \frac{4}{4x+1} - \frac{1}{x+1}$, integral $= \ln 9 - \ln 3 = \ln 3$.
12	**(e)**	$\int (1 + \frac{1}{2x}) dx = [x + \frac{1}{2}\ln x]_1^4 = 3 + \ln 2$.
13	**(c)**	I diverges (integral test, $2\sqrt{\ln x} \to \infty$); II geometric, III $p = \pi > 1$ converge.
14	**(b)**	$\sum \frac{(-1)^n}{n+1}$ converges (AST), $\sum \frac{1}{n+1}$ diverges: conditional.
15	**(c)**	Multiply by e^{2x}, $u = e^{2x}$: $\frac{1}{2}\int \frac{du}{1+u^2} = \frac{1}{2}\arctan(e^{2x}) + C$.
16	**(c)**	$u = e^x = \sin\theta$: $\int \cos^2\theta \, d\theta = \frac{1}{2}(\arcsin e^x + e^x\sqrt{1-e^{2x}}) + C$.
17	**(c)**	$R = 4$ about 3; $x = 7$ harmonic (diverges), $x = -1$ alternating (converges): $[-1, 7)$.
18	**(a)**	$\int = \ln(1+e^x) \to \infty$, so it diverges.
19	**(e)**	$P_3(x) = x + \frac{x^3}{3}$, so $\tan 1 \approx \frac{4}{3}$.
20	**(b)**	$\frac{1}{(N+1)^3} < 0.001 \Rightarrow N + 1 > 10$, so $N = 10$.